

Task 5: Malware Types & Behavior Analysis (Basic)

Tools:

Primary: VirusTotal

Objective

To understand what malware is, its types, how it behaves, how it spreads, and how it can be detected and prevented, using beginner-friendly tools.

Virus: A virus is malicious software that attaches itself to legitimate files or programs. It spreads when the infected file is executed by the user, causing system damage, data corruption, or unwanted actions without the user's knowledge.

Worm: A worm is a standalone malware that spreads automatically across networks without user interaction. It exploits security vulnerabilities to replicate itself rapidly, consuming system resources and often causing network slowdowns or failures.

Trojan: A Trojan disguises itself as legitimate or useful software to trick users into installing it. Once executed, it can steal data, create backdoors, or allow attackers remote access to the infected system.

Ransomware: Ransomware is malware that encrypts files or locks systems, making data inaccessible. Attackers demand a ransom payment, usually in cryptocurrency, in exchange for providing a decryption key to restore access.

Malware Lifecycle

The malware lifecycle consists of several stages through which malicious software operates. It begins with development, where attackers design malware for a specific purpose. Next is distribution, using emails, malicious links, or infected software. Infection occurs when the malware enters a system, followed by execution, where it performs harmful actions like data theft or encryption. Finally, persistence and impact allow malware to remain hidden, spread further, or cause long-term system damage.

How Malware Spreads

Malware spreads through various vectors, often exploiting human error and system vulnerabilities. Common methods include phishing emails with malicious attachments or links, infected websites that trigger drive-by downloads, and pirated or cracked software. Malware can also spread via USB drives, shared networks, and unpatched systems. Worms spread automatically across networks, while trojans rely on users unknowingly installing malicious programs disguised as legitimate software.

Identify Prevention Methods

Malware prevention involves a combination of technical controls and user awareness. Installing updated antivirus and anti-malware software helps detect threats early. Regular system updates and patches close security vulnerabilities. Users should avoid opening unknown email attachments or clicking suspicious links. Additional measures include firewalls, strong passwords, multi-factor authentication, and regular data backups, which reduce damage in case of ransomware or system compromise.

VirusTotal - Search - ransomware x Obtain a publicly available mal x VirusTotal - File - 34e5bd67fbd9a7040dca9cae90e36013aaeda1940bb39e7fcd5d5fa9c85cad8/detection

34e5bd67fbd9a7040dca9cae90e36013aaeda1940bb39e7fcd5d5fa9c85cad8

57 / 72
Community Score -3

57/72 security vendors flagged this file as malicious

Reanalyze Similar More

34e5bd67fbd9a7040dca9cae90e36013aaeda1940bb39e7fcd5d5fa9c85cad8
d964a8f2be0a0cfcfd085.exe

Size 256.00 KB Last Analysis Date 1 month ago

peexe long-sleeps malware calls-wmi detect-debug-environment assembly

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 10

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Popular threat label trojan.whitesnake/msil Threat categories trojan Family labels whitesnake msil mardom

Security vendors' analysis Do you want to automate checks?

AhnLab-V3	Trojan/Win.WhiteSnake.C5423097	Alibaba	TrojanPSW:MSIL/Stealer.de03c481
AliCloud	Trojan[stealer]:MSIL/WhiteSnake.C	ALYac	Gen:Trojan.Mardom.ON.21

VirusTotal - Search - ransomware x Obtain a publicly available mal x VirusTotal - File - 34e5bd67fbd9a7040dca9cae90e36013aaeda1940bb39e7fcd5d5fa9c85cad8/details

34e5bd67fbd9a7040dca9cae90e36013aaeda1940bb39e7fcd5d5fa9c85cad8

Basic properties

MD5	0a32e2ec770c67261df3f3971d517bea
SHA-1	4dfbe7f3faa5b30b9f93085572496f055b02c5e0
SHA-256	34e5bd67fbd9a7040dca9cae90e36013aaeda1940bb39e7fcd5d5fa9c85cad8
Vhash	225036755512008125a1e50b11
Authentihash	a5c46cd11e09f504d41c9f15cab299d33777d5171a67d8e255042e8f8543180
Imphash	f34d5f2d4577ed6d9ceec516c1f5a744
SSDEEP	6144:/GpIP+Qum37ZUr6dgZTp3j9bpf3X/ZZ+W0:GIPvqR7l6dgZTvfnp0
TLSH	T15C44CF0CBBC5FB5AE29E05B491A06365433492036142F78B3EDED4956FE23D05A898FF
File type	Win32 EXE executable windows win32 pe peexe
Magic	PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows
TrID	Generic CIL Executable (.NET, Mono, etc.) (67.7%) Win64 Executable (generic) (9.7%) Win32 Dynamic Link Library (generic) (6%) Win16 NE e...
DetectItEasy	PE32 Library: .NET (v4.0.30319) Linker: Microsoft Linker
Magika	PEBIN
File size	256.00 KB (262144 bytes)
PEiD packer	.NET executable

History

Creation Time	2101-02-23 00:31:07 UTC
First Submission	2023-10-01 19:01:27 UTC
Last Submission	2023-10-06 13:20:25 UTC
Last Analysis	2025-12-01 11:23:54 UTC

ByteDefend Cyber Lab flags this file as **malicious**

The provided VBA macro code is malicious and contains several suspicious actions. Here are some of the definitively malicious actions:

[Show more](#)

Hispasec flags this file as **malicious**

The macros extracted from the document exhibit several signs of malicious intent.

[Show more](#)

Popular threat label trojan.dridex/w97m **Threat categories** trojan downloader **Family labels** dridex w97m abdm

Security vendors' analysis Do you want to automate checks?

AhnLab-V3	⚠ WM/Downloader	Alibaba	⚠ TrojanDownloader:Office97/Dridex.c777...
AliCloud	⚠ Trojan.MSOffice/Downloader.aej	ALYac	⚠ Trojan.Downloader.XLS.gen
Antiy-AVL	⚠ Trojan[Downloader]/Macro.Agent	Arcabit	⚠ Trojan.Generic.D4941704
Avast	⚠ Other:Malware-gen [Trj]	AVG	⚠ Other:Malware-gen [Trj]
Avira (no cloud)	⚠ W97M/Agent.HE	BitDefender	⚠ Trojan.GenericKD.76814084

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 15

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Basic properties

MD5	1b109efade90ace7d953507adb1f1563		
SHA-1	e4eeff787b94f80e5374a2058cafc6a37d3fa2c0f		
SHA-256	2ec35da14989692e77df25d56c7b1d43a3e93a72ec2fc7cce67b2fa448744752		
Vhash	c805ed4dbfff1ad25f017d9b9a40e8cc		
SSEDPEP	384.ng/FifnjNZdOUM9zhBNHYgbY19rKNmE6DRzMMr0l2a62MOo4R/9R9Wxhu4C:9fnExagcemE6DRqmrMTMSRV9RWjc		
TLSH	T117C2C010E250F81DD73BA93DA24C14F0E12DB9819AD7628F1581F19F9BE26D7138778E		
File type	Office Open XML Spreadsheet document msoffice spreadsheet excel xlsx		
Magic	Microsoft Excel 2007+		
TrID	Excel Macro-enabled Open XML add-in (42.4%) Excel Microsoft Office Open XML Format document (with Macro) (29.2%) Excel Microsoft Office ...		
Magika	XLSX		
File size	26.61 KB (27253 bytes)		

History

Action	Date
Creation Time	2020-11-02 12:24:00 UTC
First Seen In The Wild	2021-12-07 15:22:06 UTC
First Submission	2020-11-02 12:36:30 UTC
Last Submission	2025-07-12 17:19:04 UTC

VirusTotal - Search - ransomware x Obtain a publicly available mal... x VirusTotal - File - 2ec35da1498: x +

virustotal.com/gui/file/2ec35da14989692e77df25d56c7b1d43a3e93a72ec2fc7cce67b2fa448744752/behavior

2ec35da14989692e77df25d56c7b1d43a3e93a72ec2fc7cce67b2fa448744752

Display grouped sandbox reports

BitDam ATP	1	0	0	0	0	0	C2AE	0	0	0	0	0	0
CAPE Sandbox	0	4	0	0	0	6	DOCGuard	1	2	0	0	0	0
Lastline	2	0	0	0	0	0	QiAnXin RedDrip	0	0	0	0	3	0
Tencent HABO	0	0	0	0	0	0	VMRay	1	0	0	0	8	11
Yomi Hunter	1	0	0	0	0	0	Zenbox	0	4	0	0	4	2

Activity Summary

Download Artifacts Full Reports Help

2 Detections 5 MALWARE 1 TROJAN

Mitre Signatures 1 LOW 4 INFO

IDS Rules NOT FOUND

Sigma Rules NOT FOUND

Dropped Files 1 XML 1 LNK 14 OTHER

Network comms 28 HTTP 28 DNS 29 IP 1 JA3

Upcoming Earnings

20:10 22-01-2026

VirusTotal - Search - ransomware x Obtain a publicly available mal... x VirusTotal - File - 43715401531: x +

virustotal.com/gui/file/43715401531e0060827d3dcfd406add434829192051fe76d5fddb22602cc136

43715401531e0060827d3dcfd406add434829192051fe76d5fddb22602cc136

27 / 65 Community Score

27/65 security vendors flagged this file as malicious

Reanalyze Similar More

43715401531e0060827d3dcfd406add434829192051fe76d5fddb22602cc136

Size 1.02 MB Last Analysis Date 14 days ago

4th_SOM_Meeting_Dated_24_September_2025.zip

zip

DETECTION DETAILS RELATIONS COMMUNITY 2

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Popular threat label trojan.dloadr Threat categories trojan downloader Family labels dloadr

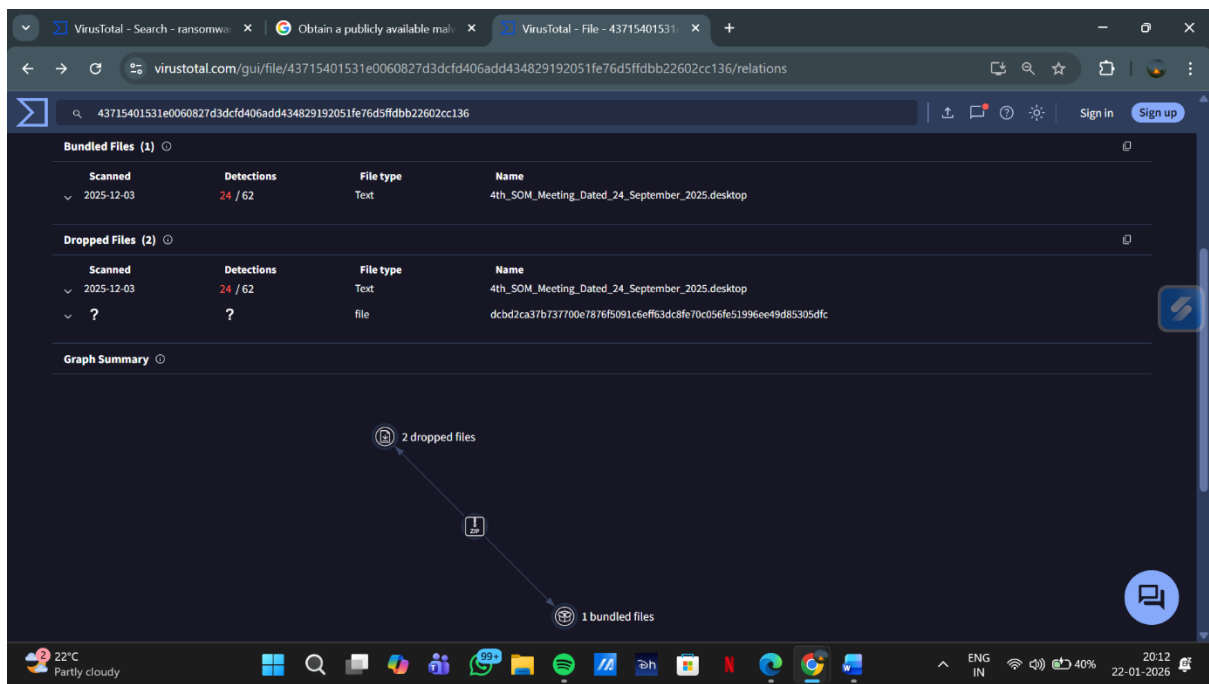
Security vendors' analysis Do you want to automate checks?

Alibaba	TrojanDownloader.Linux/Other.4ea29434	AliCloud	Trojan[downloader].Linux/Agent.SN
ALYac	Trojan.Downloader.Linux	Arcabit	Trojan.Generic.D2524CEC

22°C Partly cloudy

20:11 22-01-2026

100



Task 5 focuses on developing a foundational understanding of malware, its various types, behavior, methods of spread, lifecycle, and prevention techniques using beginner-friendly cybersecurity tools such as VirusTotal. Malware refers to malicious software designed to disrupt systems, steal data, or gain unauthorized access. Common malware types include viruses, worms, trojans, and ransomware, each differing in behavior and impact.

A virus attaches itself to legitimate files or programs and spreads when the infected file is executed by the user, often causing data corruption or system damage. Worms are standalone malware that self-replicate and spread automatically across networks without user interaction, exploiting system vulnerabilities and consuming network resources. Trojans disguise themselves as legitimate or useful software, tricking users into installation, after which they may steal sensitive information, create backdoors, or allow remote attacker access. Ransomware is particularly destructive, as it encrypts files or locks systems and demands ransom payments, usually in cryptocurrency, to restore access.

The malware lifecycle includes several stages: development, where attackers design malware; distribution through phishing emails, malicious links, or infected software; infection when malware enters a system; execution, where harmful activities such as data theft or encryption occur; and persistence and impact, allowing malware to remain hidden, spread further, or cause long-term damage.

Malware spreads through multiple vectors including phishing emails, infected websites, pirated software, USB drives, shared networks, and unpatched systems. Worms spread automatically, while trojans rely heavily on user interaction.

Preventing malware requires both technical controls and user awareness. Key prevention methods include installing updated antivirus software, applying regular system patches, avoiding suspicious links and attachments, using firewalls, implementing strong passwords and multi-factor authentication, and maintaining regular data backups. Together, these measures significantly reduce the risk and impact of malware attacks.